

1a. The Unit Step Function (Heaviside Function)

In engineering applications, we frequently encounter functions whose values change abruptly at specified values of time t . One common example is when a voltage is switched on or off in an electrical circuit at a specified value of time t .

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[Shifted unit step function](#)

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The value of $t = 0$ is usually taken as a convenient time to switch on or off the given voltage.

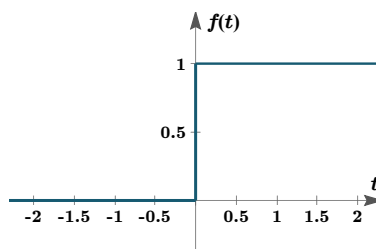
The switching process can be described mathematically by the function called the **Unit Step Function** (otherwise known as the **Heaviside function** after [Oliver Heaviside](#)).

The Unit Step Function

Definition: The unit step function, $u(t)$, is defined as

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

That is, u is a function of time t , and u has value **zero** when time is negative (before we flip the switch); and value **one** when time is positive (from when we flip the switch).



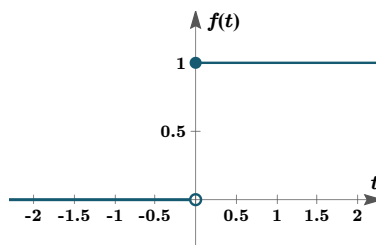
Graph of $f(t) = u(t)$, the unit step function.

Value at $t = 0$?

In some text books you will see the unit step function defined as having value 1 at $t = 0$, as follows:

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

We would indicate the discontinuity on our graph like this:



Graph of $f(t) = u(t)$, the unit step function, with $f(0) = 1$.

Also, sometimes you'll see the value given as $f(0) = 0.5$.

In this work, it doesn't make a great deal of difference to our calculations, so we'll continue to use the first interpretation, and draw our graphs accordingly.

Shifted Unit Step Function

In many circuits, waveforms are applied at specified intervals other than $t = 0$. Such a function may be described using the **shifted** (aka **delayed**) unit step function.

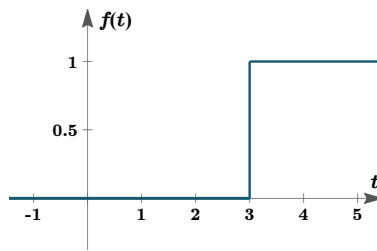
Definition of Shifted Unit Step Function

A function which has value 0 up to the time $t = a$ and thereafter has value 1, is written:

$$u(t - a) = \begin{cases} 0 & \text{if } t < a \\ 1 & \text{if } t \geq a \end{cases}$$

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The sketch of the waveform is as follows:



Graph of $f(t) = u(t - 3)$, a shifted unit step function.

Rectangular Pulse

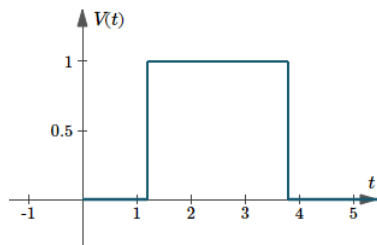
A common situation in a circuit is for a voltage to be applied at a particular time (say $t = a$) and removed later, at $t = b$ (say). We write such a situation using unit step functions as:

$$V(t) = u(t - a) - u(t - b)$$

This voltage has strength 1, duration $(b - a)$.

Example 2 - Rectangular Pulse

The graph of $V(t) = u(t - 1.2) - u(t - 3.8)$ is as follows. Here, the duration is $3.8 - 1.2 = 2.6$.



Graph of $V(t) = u(t - 1.2) - u(t - 3.8)$, an example of a rectangular pulse.

Exercises

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Write the following functions in terms of **unit step** function(s). Sketch each waveform.

(a) A 12-V source is switched on at $t = 4$ s.

[Answer](#)

(b) $V(t) = \begin{cases} 1 & 0 < t < a \\ 0 & t > a \end{cases}$

(Assume $a > 0$.)

[Answer](#)

(c) One cycle of a square wave, $f(0) = 4$, amplitude = 4, period = 2 seconds.

[Answer](#)

(d) The unit Ramp function (i.e. $f(t) = t$ for $t > 0$)

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[Answer](#)

$$(f) V(t) = \begin{cases} 0 & t < 3 \\ 2t + 8 & 3 < t < 5 \\ 0 & t > 5 \end{cases}$$

[Answer](#)

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